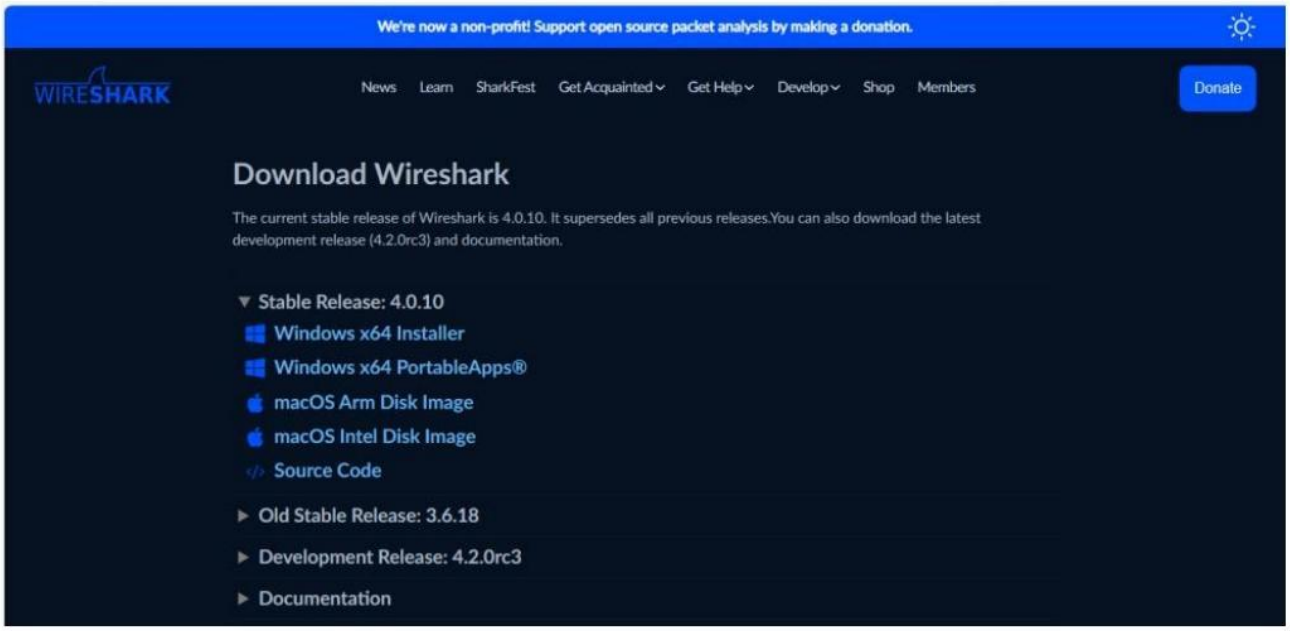
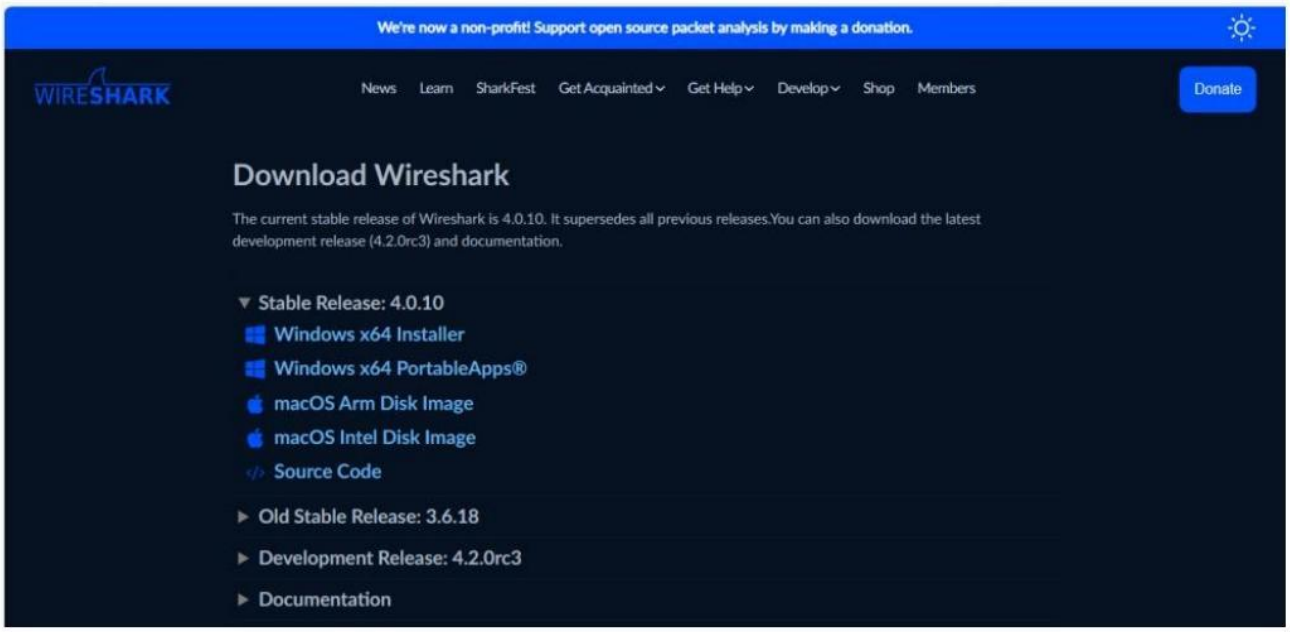
# Network test tools and network design simulation (individual assignment)

## 0. Environment

- Ubuntu 20.04 LTS (recommended)
- Wireshark 4.0.10 you can download it from this URL: https://www.wireshark.org/download.html




- Cisco Packet Tracer 8.2.1 you can download it from this URL: https://skillsforall.com/resources/lab-downloads



1. **Network Test Tool Use – Linux basic network test commands and wireshark packet capture experiment 【50 points, complete it alone】**

   A . [10 points] Be proficient in using commands such as ifconfig, ping, nslookup, arp, netstat,traceroute, etc., and try to explain what protocol they are all done with.

   B . [10 points] Capture packets and parse the complete process of TCP three-way handshake and four-way handshake, the characteristics of UDP data transmission, and explain the reasons for protocol selection in combination with actual application scenarios (such as using UDP for DNS and TCP for HTTP).

   C . [20 points] Use the wireshark (GUI) and the Tshark (command) packet capture tool to grab ARP, ICMP, DNS, HTTP, TCP, UDP, etc., and parse the packet information content in the packet. (The contents of the bag you caught may not be the same as what you learned, please explain why?)

   D . [10 points]Track the encapsulation process of an HTTP request from the application layer to the physical layer, as well as the unencapsulation process of the response packet, and label the core information of the header fields at each layer.

2. **Network Design Simulation – Use Cisco Packet Tracer to complete network simulation experiment 【50 points, complete it alone】**

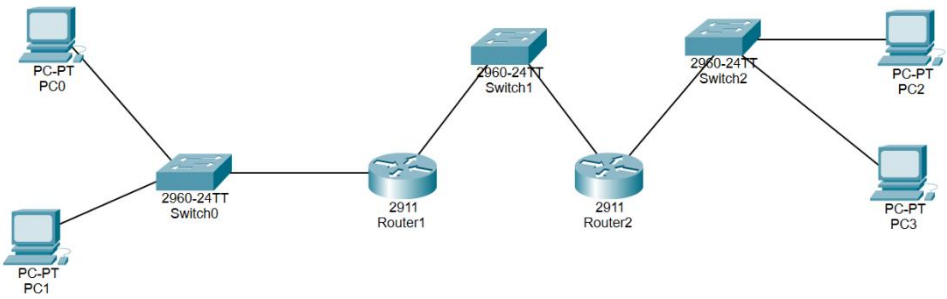
   Step 1: use Cisco Packet Tracer to complete the network topology design and configure the network.

   Step 2: network connectivity verification and conclusion analysis.

   Follow the preceding two steps to complete the network topology shown in the following 2 tasks.

   Please write about the process you completed and the results of your tests in your report named by name id-simulation.pdf, and you are encouraged to include any difficulties you encountered. (20 points)

   **Task 1**

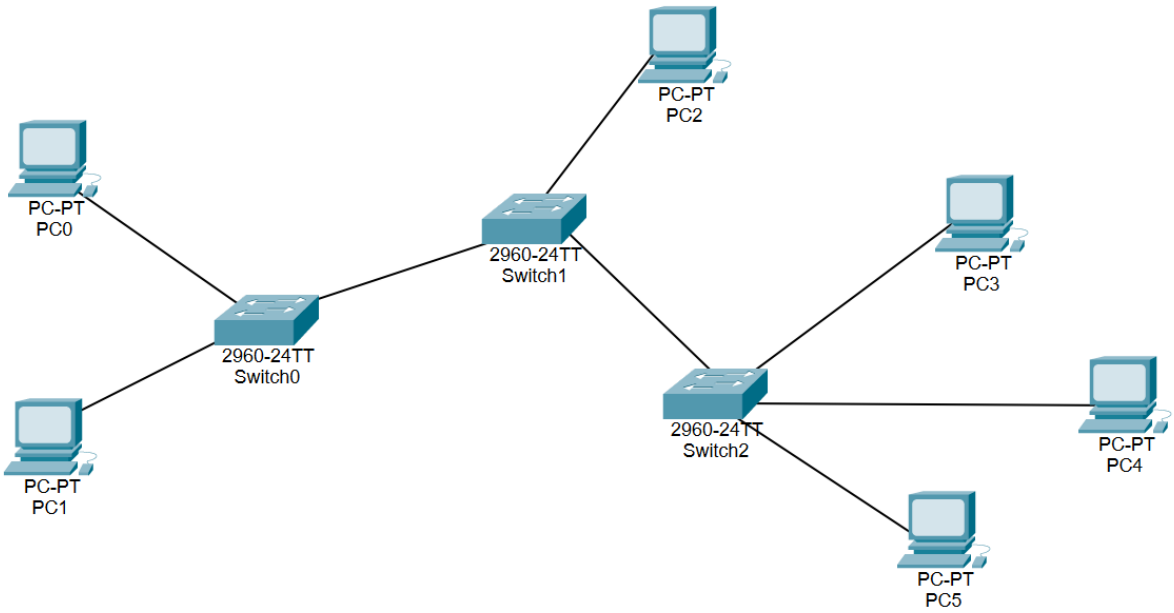



| device | IP |
|---|---|
| **Router1** | **Interface 0: 192.168.1.1/24**<br>**Interface 1: 192.168.2.1/24** |
| **Router2** | **Interface 0: 192.168.2.2/24** |

| | |
|---|---|
| | **Interface 1: 192.168.3.1/24** |
| **PC 0** | **192.168.1.10/24** |
| **PC 1** | **192.168.1.11/24** |
| **PC 2** | **192.168.3.10/24** |
| **PC 3** | **192.168.3.11/24** |

Please save the file as name-id-task1.pkt (10 points). Ensure that your configuration is completed successfully and that we can assess successfully. Please write your name and student number in the blank areas of the packet tracer and make sure they are saved together in the .pkt file

**Task 2**




| **Device** | **VLAN** | **IP** |
|---|---|---|
| **Switch0** | **10** | **-** |
| **Switch1** | **20** | **-** |
| **Switch2** | **30** | **-** |
| **PC0** | **10** | **192.168.10.10/24** |
| **PC1** | **10** | **192.168.10.11/24** |
| **PC2** | **20** | **192.168.20.10/24** |
| **PC3** | **30** | **192.168.30.10/24** |

| | | |
|---|---|---|
| **PC4** | **30** | **192.168.30.11/24** |
| **PC5** | **30** | **192.168.30.12/24** |

**VLANs are helpful in the administration of logical groups, allowing members of a group to be easily moved, changed or added. This activity focuses on creating and naming VLANs and assigning access ports to specific VLANs.**

Please save the file as name-id-task2.pkt (20 points). Ensure that your configuration is completed successfully and that we can assess successfully. Please write your name and student number in the blank areas of the packet tracer and make sure they are saved together in the .pkt file

3. **Reference**

   https://www.netacad.com/courses/packet-tracer (it is highly recommended that you learn how to use the Cisco Packet Tracer via this URL)

4. **Submit**

   - **Due on 23:59, 14 Dec. 2025**
   - **Please compress all files and name as name-id.zip**
   - **Cheating or abuses will not be tolerated Due on 23:59, 20 Dec. 2025**